

Course: B.Tech.

Semester : IV

Branch : Artificial Intelligence & Data Science / Artificial Intelligence(AI) & Data Science

Subject Code & Name: BTAIC401 – Data Analysis

Max Marks: 60

Date: 19/12/2024

Duration: 3 Hrs.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

	(Level/ CO)	Marks
Q. 1 Objective type questions. (Compulsory Questions)		(12)
1 Statistics is the a. fun of finding patterns in data b. pleasure of making discoveries c. the ability to guide decisions d. All of the mentioned	Remember	1
2 Which of the following is not a parameter for Measurement scale? a. Nominal b. Time c. Ordinal d. Cyclic	Analysis	1
3 Data Analysis is defined by the statistician? a. William S. b. Hans Peter Luhn c. Gregory Shapiro d. John Tukey	Remember	1
4 Data Analysis is a process of _____ data? a. inspecting b. cleansing c. transforming d. All of the mentioned	Remember	1
5 Which of the following is not a major data analysis approach? a. Data Mining b. Predictive Intelligence c. Business Intelligence d. Text Analytics	Apply	1
6 Consider a Situation where a researcher wants to test whether the populations mean is lower than some hypothesized value, then the researcher would employ _____ test. a. Z-tailed b. Left tailed c. Right tailed d. F-test	Evaluating	1
7 While performing Data Analysis, which of the following graphical methods of representing data would be useful for showing trends? a. Bar charts b. Line charts c. Box plots d. Pie charts	Evaluating	1
8 Which of the following is a data visualization method? a. Line b. Circle and Triangle c. Pie chart and Bar chart d. Pentagon	Analysis	1

9	What is the primary goal of Data Analysis?	Remember	1
	a. making predictions b. Data visualization c. Designing databases d. Summarizing and interpreting data		
10	How is the range of a dataset calculated?	Analysis	1
	a. sum of all values b. product of all values c. average of all values d. The difference between the largest and smallest values		
11	How many main statistical methodologies are used in data analysis?	Analysis	1
	a. 2 b. 3 c. 4 d. None of the mentioned		
12	Graphical representation of a frequency distribution is called as	Remember	1
	a. Bar chart b. Line chart c. Histogram d. Pie chart		

Q. 2 Solve the following. (12)

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|----|--|----------|---|
| A) | Explain several "bias" in statistical contexts? | Remember | 6 |
| B) | Explain PPDAC as an iterative statistical process? | Apply | 6 |

Q.3 Solve the following. (12)

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|----|---|------------|---|
| A) | Explain the importance of data preparation and data cleaning? | Evaluating | 6 |
| B) | Explain several measures of central tendency in descriptive statistics? | Remember | 6 |

Q. 4 Solve Any Two of the following. (12)

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|----|---|----------|---|
| A) | Explain Logit curve and Logistic curve with neat diagram? | Remember | 6 |
| B) | Explain Box-Cox transform with a suitable example? | Analysis | 6 |
| C) | Explain Log and exponential transforms with a suitable example? | Analysis | 6 |

Q.5 Solve Any Two of the following. (12)

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|----|--|----------|---|
| A) | Explain McNemar's test by considering an example? | Analysis | 6 |
| B) | Define Chi-Square Test? Mention few Applications of Chi-Square test? | Analysis | 6 |
| C) | Write short note on "Goodness of fit tests"? | Remember | 6 |

Q. 6 Solve Any Two of the following. (12)

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|----|--|----------|---|
| A) | Explain the steps to perform Mood's Median test with an example? | Apply | 6 |
| B) | Explain Kruskal Wallis ANOVA with an example? | Remember | 6 |
| C) | Define ANOVA? Explain One way ANOVA with examples? | Applying | 6 |

*** End ***